

TELCO CUSTOMER CHURN PREDICTION

Modular Architecture Diagram

Author: UPPALA VENKATA SATYA SRINIVAS | 2025



MODULE INTERFACES & ARTIFACT FLOW

Inputs, Outputs, and Saved Artifacts per Module

Module	Key Inputs	Key Outputs	Saved Artifact
1. Data Ingestion	Raw CSV (7,043 rows)	Clean DataFrame 7,032 rows	—
2. EDA	Clean DataFrame	Charts & Stats Insights report	—
3. Feature Eng.	Clean DataFrame	X_train, X_test y_train, y_test	scaler.pkl feature_columns.pkl
4. SMOTE + Train	X_train (scaled) y_train	Trained XGBoost SMOTE balanced	churn_xgb_model.pkl
5. Evaluation	Model + X_test y_test	Metrics, ROC Feature Importance	correct_churn_ predictions.csv
6. Deployment	All .pkl artifacts Raw customer data	Churn probability Batch predictions	app.py .pbix dashboard

TECHNOLOGY STACK

- Python 3.x**
Core language for all pipeline stages
- Scikit-Learn**
Preprocessing, Logistic Reg, Random Forest, SMOTE
- imbalanced-learn**
SMOTE for synthetic minority oversampling
- Power BI**
Business intelligence dashboard
- Pandas / NumPy**
Data cleaning, transformation, EDA
- XGBoost**
Final classification model (best performance)
- Streamlit**
Web application for real-time churn prediction
- Matplotlib/Seaborn**
Visualizations: EDA charts, ROC curve, feature importance